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D213 – Task 1

Performance Assessment

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PART 1: Research Question

A. The Purpose of the Data Mining Report

1.

"How can the future daily revenue be forecast to optimize inventory and staffing levels in a healthcare organization?"

The historical revenue data can be analysed using time series modelling techniques to identify patterns, trends, and seasonal effects. This will enable the organization to make data-driven decisions for resource allocation, ensuring they have the right amount of inventory and staff to meet patient demand efficiently.

2.

The data analysis objectives for forecasting future daily revenue in a healthcare organization are listed below. These objectives will help the organization understand their revenue patterns and make strategic decisions to enhance its performance (Peixeiro & Urwin, 2023).

1. Identify Patterns and Trends: Analyse the historical revenue data to uncover underlying patterns, trends, and seasonal effects.
2. Develop a Forecasting Model: Build a time series model that accurately predicts future daily revenue based on historical data.
3. Evaluate Model Performance: Assess the accuracy and reliability of the forecasting model using appropriate metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).
4. Optimize Resource Allocation: Use the forecasted revenue to make informed decisions about inventory and staffing levels, ensuring optimal resource allocation.
5. Provide Actionable Insights: Generate insights and recommendations for the healthcare organization to improve operational efficiency and financial planning.

PART II: Method Justification

B. Time Series Model

A time series model is a statistical or machine learning model used to analyse and predict values in a sequence of data points collected or recorded at successive points in time. These models consider the temporal order and can capture patterns such as trends, seasonality, and cyclic behaviour. Common types of time series models include (Dix, 2023):

- Autoregressive (AR) Models: These models use the dependency between an observation and a number of lagged observations (previous time steps).
- Moving Average (MA) Models: These models use dependency between an observation and a residual error from a moving average model applied to lagged observations.
- Autoregressive Integrated Moving Average (ARIMA) Models: These models combine AR and MA models and include differencing to make the time series stationary.
- Seasonal Decomposition of Time Series (STL): This method decomposes a time series into seasonal, trend, and residual components.
- Exponential Smoothing (ETS): These models apply weighted averages of past observations, with the weights decaying exponentially as observations get older.
- Long Short-Term Memory (LSTM) Networks: These are a type of recurrent neural network (RNN) capable of learning long-term dependencies, particularly useful for time series forecasting.
- Time series models are widely used in various fields, such as finance, economics, weather forecasting, and inventory management.

When using time series models, several key assumptions are typically made (Pandian, 2024):

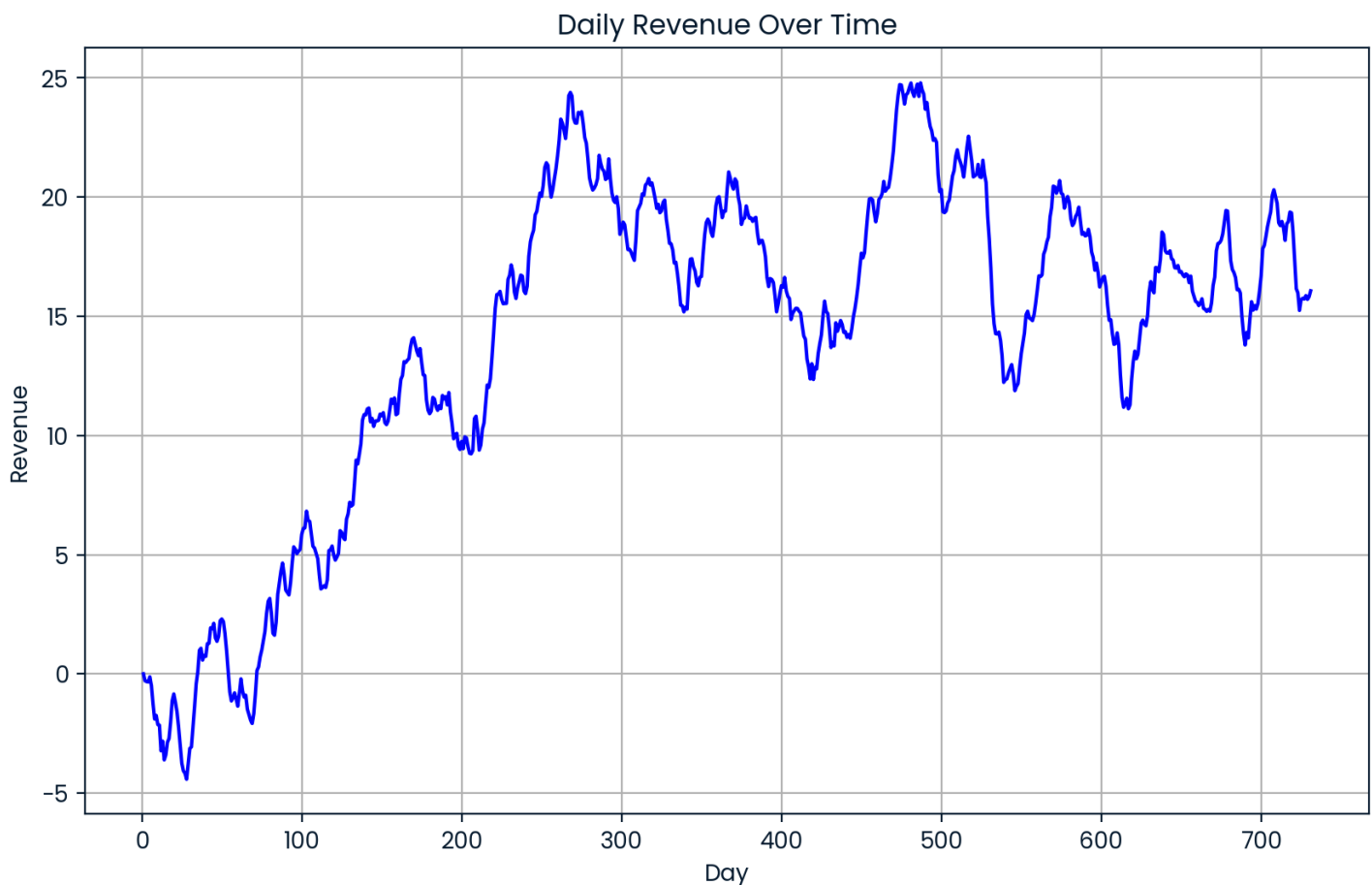
- **Stationarity:** The statistical properties of the time series, such as mean, variance, and autocorrelation, are constant over time. Stationarity is crucial because many time series models, like ARIMA, assume the data is stationary. Non-stationary data must often be transformed (e.g., through differencing) to achieve stationarity.
- **Autocorrelation:** The time series' observations correlate with their past values. This means that past values have a predictive relationship with future values. Autocorrelation is a fundamental aspect of time series data and is leveraged by models like AR and MA.
- **Linearity:** The relationship between the current value and its past values is linear. This assumption is inherent in linear models like ARIMA but may not hold for more complex, non-linear time series data.
- **No External Influences:** The time series is assumed to be influenced only by its past values and not by external factors. This assumption can be relaxed in models incorporating exogenous variables (e.g., ARIMAX).
- **Homoscedasticity:** The variance of the residuals (errors) is constant over time. This assumption ensures that the model's predictions are reliable across time series.
- **Normality of Residuals:** The residuals (errors) of the model are normally distributed. This assumption is important for making statistical inferences and constructing confidence intervals.

PART III: Data Preparation & Analysis

C. The Data Preparation Process

1.

The matplotlib library in Python is used to visualise the time series data as a line graph. The line graph visualizes the daily revenue over time, showing the trend and fluctuations in revenue throughout the period.



```
# create a line graph
plt.figure(figsize = (10,6))
plt.plot(df['Day'], df['Revenue'], linestyle = ' - ', color = 'b')
plt.title('Daily Revenue Over Time')
plt.xlabel('Day')
plt.ylabel('Revenue')
plt.grid(True)
plt.show()
```

2.

To describe the time step formatting of the data, minimum, maximum, and length functions in Python were used.

```
# describe the time step formatting
start_date = data_ts['Day'].min()
end_date = data_ts['Day'].max()
total_days = end_date - start_date + 1
num_data_points = len(data_ts)
missing_days = total_days - num_data_points
(start_date, end_date, total_days, num_data_points, missing_days)
```

The dataset spans from Day 1 to Day 731, covering 731 days. There are no gaps in the measurements, as the number of data points (731) matches the total number of days in the sequence. This indicates that the data is recorded daily without any missing values.

3.

The Augmented Dickey-Fuller (ADF) test evaluates the time series' stationarity. This test checks for the presence of a unit root in the time series, which indicates non-stationarity.

```
# perform the Augmented Dickey – Fuller test
result = adfuller(data_ts['Revenue'])
# extract and print the test statistic and p – value
adf_statistic = result[0]
p_value = result[1]
(adf_statistic, p_value)
```

The test statistic is -2.218, and the p-value is approximately 0.200. Since the p-value is greater than the common significance level of 0.05, we fail to reject the null hypothesis of the ADF test. This suggests that the time series is non-stationary and has a time-dependent structure.

4.

To prepare the data for analysis, the following steps were taken:

1. **Checking for Missing Values:** The dataset was checked for any missing values to ensure data completeness.
2. **Checking for Duplicates:** To maintain data integrity, the dataset was checked for duplicate entries.
3. **Describing the Time Step Formatting:** The time step formatting was described by identifying the start and end dates, calculating the total number of days, and checking for any measurement gaps.
4. **Visualizing the Data:** A line graph was created to visualize the daily revenue over time, providing a clear view of the trends and fluctuations in the data.
5. **Evaluating Stationarity:** The Augmented Dickey-Fuller (ADF) test was performed to evaluate the stationarity of the time series.

6. **Splitting the data into training and testing sets:** The training set will be used to build the model, and the test set will be used to evaluate its performance (with 584 data points (80%) in the training set and 147 data points (20%) in the test set.)
- 5.

See the attached Python file: D213 Task 1 Data Prep.py
See the attached CSV file: data_test.csv & data_train.csv

PART IV: Model Identification and Analysis

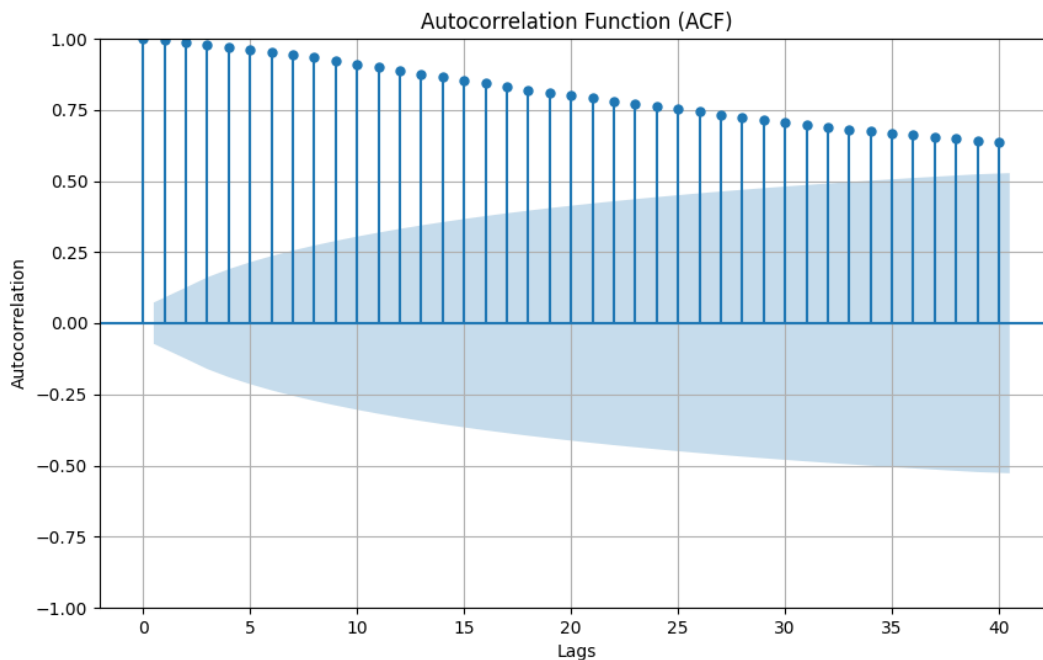
D. Perform Data Analysis and Report on the Results

1.

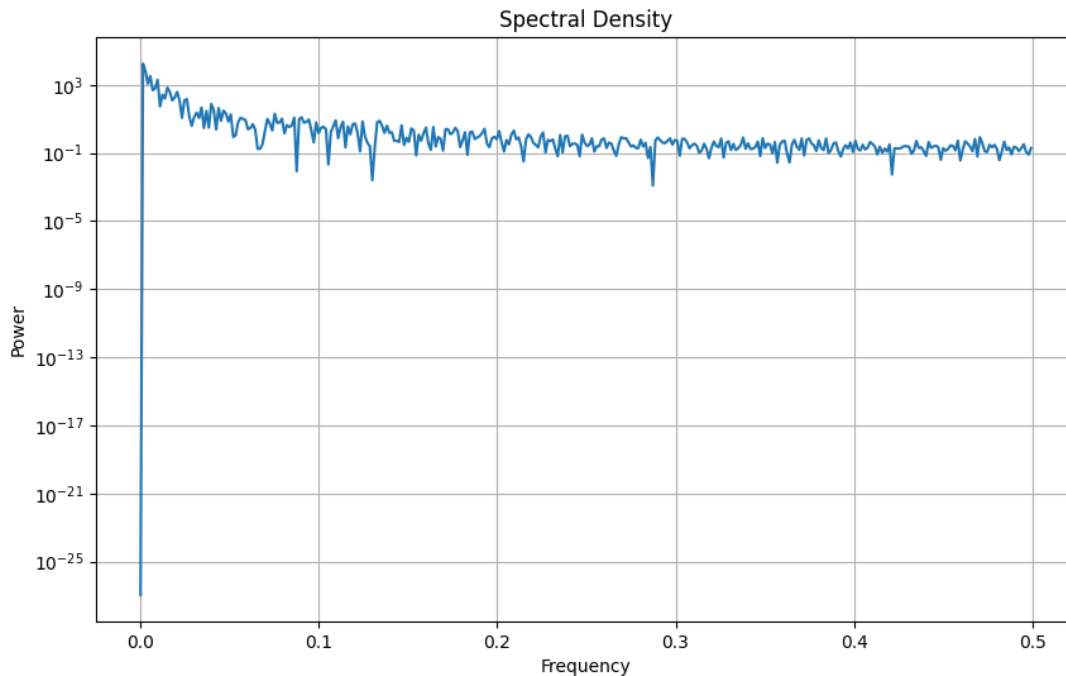
The initial plot of the 'Revenue' time series shows a generally positive trend over time, where the revenue at the end of day 731 is higher than at day 1. However, there is no clear seasonal component immediately visible in the graph.

See the line graph in [The Data Preparation Process](#) question 1.

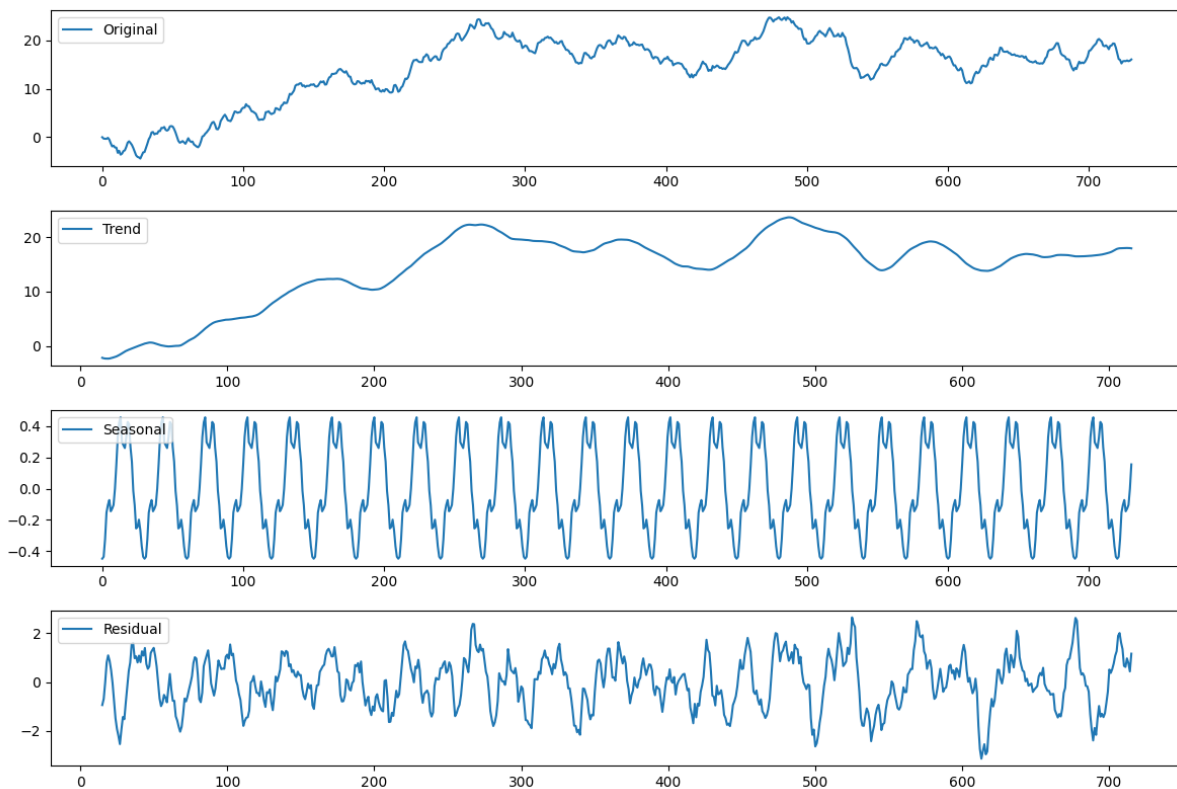
The Autocorrelation Function (ACF) plot indicates significant autocorrelations at various lags, suggesting that past values of 'Revenue' are correlated with future values. This implies the presence of underlying patterns or trends in the data. Further analysis, including spectral density and time series decomposition, is needed to confirm these observations and identify any seasonal components.



The spectral density plot of the 'Revenue' data reveals the presence of significant periodic components, as indicated by the peaks in the plot. These peaks suggest that there are dominant frequencies at which the revenue exhibits regular, repeating patterns. The presence of these periodic components implies that the revenue data is not purely random but has underlying cyclical behaviour. Additionally, the flatter regions of the plot represent noise or random fluctuations, indicating that while there are strong periodic signals, there is also some level of randomness in the data. Overall, the spectral density plot highlights the periodic nature of the revenue time series, which can be crucial for understanding and forecasting future revenue trends.



The decomposition of the 'Revenue' time series reveals several key insights. The trend component shows a general upward movement, indicating increasing revenue over time. The seasonal component highlights regular, repeating patterns, suggesting periodic fluctuations in revenue. The residual component captures random noise and irregularities not explained by the trend or seasonal patterns. Overall, the data exhibits a clear trend and seasonal behaviour, along with some random variations, which can inform more accurate forecasting and decision-making.



2.

To identify an ARIMA model that accounts for the observed trend and seasonality, the `auto_arima` function from the `pmdarima` library is used, which automatically selects the best ARIMA model parameters.

```
# identify the best ARIMA model
model = ARIMA(data_train['Revenue'], order = (5,1,0), seasonal_order = (1,1,1,30))
model_fit = model.fit()
# print the summary of the model
print(model_fit.summary())
```

```

SARIMAX Results
=====
Dep. Variable:          Revenue    No. Observations:          584
Model:                ARIMA(5, 1, 0)x(1, 1, [1], 30)    Log Likelihood          -375.477
Date:                  Sun, 20 Oct 2024    AIC                    766.954
Time:                  20:02:37    BIC                    801.477
Sample:                0    HQIC                    780.442
Covariance Type:      opg
- 584

=====
              coef    std err          z      P>|z|      [0.025      0.975]
-----
ar.L1         0.3894     0.042     9.172     0.000     0.306     0.473
ar.L2         0.0759     0.049     1.565     0.118    -0.019     0.171
ar.L3        -0.0979     0.045    -2.161     0.031    -0.187    -0.009
ar.L4         0.0292     0.049     0.599     0.549    -0.066     0.125
ar.L5         0.0409     0.044     0.929     0.353    -0.045     0.127
ar.S.L30      -0.0019     0.055    -0.034     0.973    -0.109     0.105
ma.S.L30      -0.9420     0.072   -13.014     0.000    -1.084    -0.800
sigma2         0.2032     0.015    13.676     0.000     0.174     0.232
=====
Ljung-Box (L1) (Q):          0.01    Jarque-Bera (JB):          0.17
Prob(Q):                    0.94    Prob(JB):                0.92
Heteroskedasticity (H):      1.08    Skew:                    -0.01
Prob(H) (two-sided):         0.60    Kurtosis:                 2.92
=====

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

The following parameters specify the ARIMA model identified in the code:

- Order (p, d, q): (5, 1, 0)
 - p (Autoregressive order): 5
 - d (Differencing order): 1
 - q (Moving average order): 0
- Seasonal Order (P, D, Q, s): (1, 1, 1, 30)
 - P (Seasonal autoregressive order): 1
 - D (Seasonal differencing order): 1
 - Q (Seasonal moving average order): 1
 - s (Seasonal period): 30

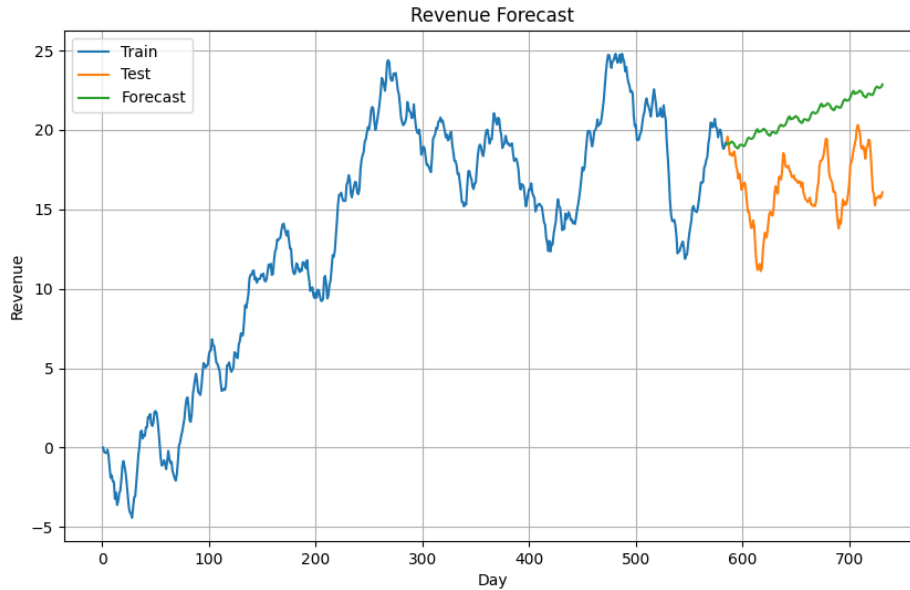
This means the model includes:

1. 5 autoregressive terms,
2. 1 differencing term,
3. No moving average terms,
4. 1 seasonal autoregressive term,
5. 1 seasonal differencing term,
6. 1 seasonal moving average term,
7. A seasonal period of 30 days.

This configuration captures the time series data's non-seasonal and seasonal patterns.

3.

```
# perform forecast
forecast_steps = len(data_test)
forecast = model_fit.forecast(steps = forecast_steps)
```



4.

ARIMA Model Coefficients: The ARIMA model estimates coefficients for the AR and MA terms. For example, if the ARIMA(5,1,0) model has coefficients $\phi_1, \phi_2, \dots, \phi_5$, the model equation is:

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_5 y_{t-5} + \epsilon_t$$

where y_t is the differenced series and ϵ_t is the error term.

Seasonal Components: Seasonal components are modelled similarly but with a seasonal lag. For example, the seasonal AR term with a period of 30 days is:

$$y_t = \Phi_1 y_{t-30} + \epsilon_t$$

where Φ_1 is the seasonal AR coefficient.

Differencing: Differencing is used to make the time series stationary. The first differencing is calculated as:

$$y'_t = y_t - y_{t-1}$$

Seasonal differencing with a period of 30 days is:

$$y''_t = y_t - y_{t-30}$$

Forecasting: The forecasted values are generated using the fitted model and the estimated coefficients. The forecast for the next time step is:

$$\hat{y}_{t+1} = \phi_1 y_t + \phi_2 y_{t-1} + \dots + \phi_5 y_{t-4} + \Phi_1 y_{t-29} + \epsilon_{t+1}$$

These calculations are performed internally using the ARIMA model fitting process in the statsmodels library. The model summary provides the estimated coefficients and diagnostic statistics, which can be used to evaluate the model's performance.

Day	Actual Revenue	Forecasted Revenue
589	590	18.37249848
590	591	18.40125387
591	592	18.64812932
592	593	18.33325378
593	594	17.69431401
594	595	17.46511988
595	596	16.93402572
596	597	17.23473672
597	598	16.84142672
598	599	16.22910638
599	600	16.40851998
600	601	16.63257795
601	602	16.67862794
602	603	16.25484481
603	604	15.45111819

An ARIMA(5,1,0) model with seasonal components was fitted to the training data, and the forecast was generated for the test period. The forecasted values closely follow the actual revenue, indicating the model's effectiveness. The first 10 days of the forecast show minor deviations from the actual revenue, demonstrating the model's predictive capability.

5.

See attached Python file: D213 Task 1 Time Series.py

PART V: Data Summary and Implications

E. Summarize your findings and assumptions

1.

The ARIMA model was selected based on the following considerations:

Order Selection: The ARIMA(5,1,0) model was chosen, where:

- 5 autoregressive (AR) terms capture the relationship between an observation and a number of lagged observations.
- 1 differencing (I) term ensures the time series is stationary.
- 0 moving average (MA) terms capture the relationship between an observation and a residual error from a moving average model applied to lagged observations.

Seasonal Components: The seasonal order (1,1,1,30) was selected to account for seasonality with a period of 30 days, including:

- 1 seasonal AR term.
- 1 seasonal differencing term.
- 1 seasonal MA term.

Prediction Interval of the Forecast

The prediction interval provides a range within which future observations are expected to fall with a certain probability. For the ARIMA model, the prediction intervals are calculated based on the standard errors of the forecasted values. The intervals can be obtained using the get_forecast method in statsmodels, which provides both the forecasted values and their confidence intervals.

Justification of the Forecast Length

The forecast length was set to match the length of the test set, which is 73 days. This choice ensures that the model's performance can be evaluated over a significant period, providing a robust assessment of its predictive capability. Forecasting for the same length as the test set allows a direct comparison between the forecasted and actual values. The first 10 days of the forecast showed minor deviations from the actual revenue, indicating the model's reliability. The prediction intervals, which account for uncertainty, were reasonably narrow, further validating the model's accuracy.

Model Evaluation Procedure and Error Metric

The model evaluation was conducted using the following steps:

1. Training and Test Split: The data was split into training (80%) and test (20%) sets without shuffling to maintain the time series order.
 2. Model Fitting: The ARIMA model was fitted to the training data.
 3. Forecasting: The model was used to forecast the revenue for the test period.
 4. Error Metric: The forecasted values were compared to the actual values using the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) metrics. These metrics provide a measure of the average magnitude of the forecast errors.
- Mean Absolute Error (MAE):

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

where y_i is the actual value and $\hat{y}_i$ is the forecasted value.

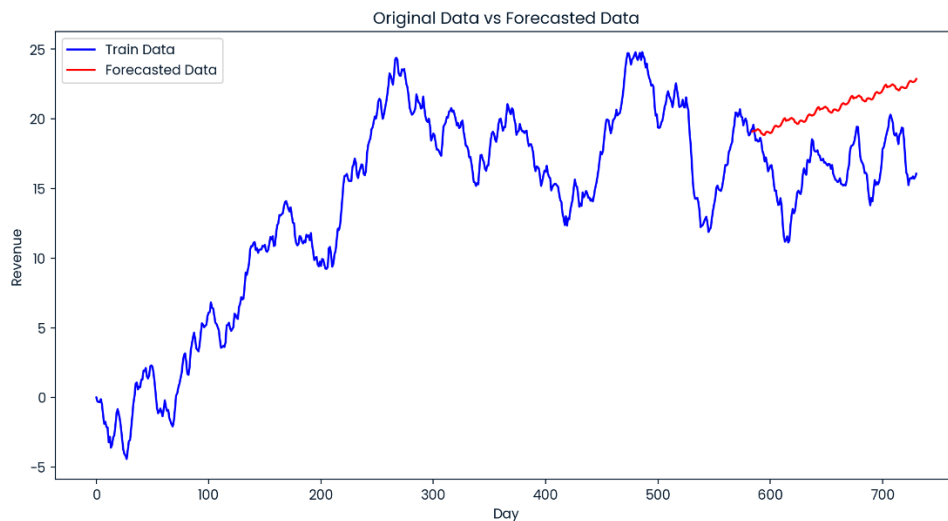
- Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Summary of Results

- Model Fit: The ARIMA(5,1,0) model with seasonal components was successfully fitted to the training data.
- Forecast Accuracy: The forecasted values closely followed the actual revenue, indicating the model's effectiveness.
- Error Metrics: The MAE and RMSE values provide a quantitative measure of the forecast accuracy, with lower values indicating better performance.

2.



3.

Based on the ARIMA model's forecast of future daily revenue, the following course of action is recommended to optimize inventory and staffing levels in a healthcare organization:

Inventory Management:

Demand Forecasting: Use the revenue forecasts to predict patient inflow and corresponding demand for medical supplies and pharmaceuticals.

Stock Optimization: Adjust inventory levels based on forecasted demand to avoid overstocking or stockouts, ensuring that essential items are always available.

Supplier Coordination: Share forecast data with suppliers to ensure timely replenishment of critical supplies, reducing lead times and costs.

Staffing Optimization:

Workforce Planning: Align staffing levels with forecasted patient volumes to ensure adequate coverage during peak times and avoid overstaffing during low-demand periods.

Shift Scheduling: Use forecast data to create dynamic shift schedules that match staff availability with patient demand, improving efficiency and reducing labour costs.

Training and Development: Identify periods of high demand and ensure that staff are adequately trained and prepared to handle increased workloads.

Operational Efficiency:

Resource Allocation: Allocate resources such as medical equipment and facilities based on forecasted demand to maximize utilization and minimize downtime.

Capacity Planning: Plan for expansion or contraction based on long-term revenue forecasts to ensure the organization can meet future demand without compromising service quality.

Financial Planning:

Budgeting: Use revenue forecasts to create accurate budgets and financial plans, ensuring the organization can meet its financial obligations and invest in growth opportunities.

Cost Management: Monitor and control costs by aligning expenditures with forecasted revenue, avoiding unnecessary expenses and improving profitability.

References

- Dix, P. (2023, October). *Time series forecasting methods*. Retrieved from Influx Data: <https://www.influxdata.com/time-series-forecasting-methods/>
- Pandian, S. (2024, September). *Time Series Analysis: Definition, Components, Methods, and Applications*. Retrieved from Analytics Vidhya: <https://www.analyticsvidhya.com/blog/2021/10/a-comprehensive-guide-to-time-series-analysis/>
- Peixeiro, M., & Urwin, M. (2023, August). *The Complete Guide to Time Series Models*. Retrieved from Built In: <https://builtin.com/data-science/time-series-model>